Greta Ru-Mei Zu

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EDUCATION

Imperial College London — London, UK
MSc Computing (Software Engineering) — *Sep 2026 – Sep 2027 (expected)*

McGill University — Montreal, Canada
BSc Computer Science (Artificial Intelligence) — *Aug 2022 – May 2026*

- **GPA: 3.85 / 4.0**; Alma Mater Scholarship (2022)
- Relevant coursework: Algorithms & Data Structures, Applied Machine Learning, Artificial Intelligence, Reinforcement Learning, Data Science, Distributed Systems, Operating Systems, Compiler Design, Database Systems

EXPERIENCE

Ericsson Canada — Montreal, Canada
AI Software Developer Intern — *Jan 2026 – May 2026*

- Built and deployed a **retrieval-augmented generation (RAG)** chatbot using Hopsworks, internal EricAI models, and DeepEval for automated LLM evaluation.
- Extended CANDI, an internal network-dimensioning tool, with ML capabilities and automated the build of its prediction model.
- Shipped a price-prediction tool combining regex/LLM data extraction with a trained **LightGBM** forecasting model.

Machine Learning Intern — *May 2025 – Dec 2025*

- Designed and deployed serverless RAG and preprocessing pipelines on **AWS** (Lambda, Bedrock, S3).
- Built Python agentic workflows that generate fine-tuning datasets, improving latency and output quality by ~**40%**.
- Developed a CLI coding-assistant tool with LangChain DeepAgents.

RESEARCH

EPFL – Scalable Computing Systems (SaCS) Lab — Lausanne, Switzerland
Summer@EPFL Research Intern · Prof. Anne-Marie Kermarrec — *Jun 2026 – Aug 2026*

- Designed and implemented a free-rider attack in a **federated learning** context, that evades detection while doing only ~**30%** of an honest client's work, showing these defences are fragile.
- Contributed experimental and theoretical results to the paper "Head-Only Watermarking Cannot Stop Free-Riders in Federated Learning".

McGill University – DISC Lab — Montreal, Canada
Undergraduate Researcher · Profs. Oana Balmau & Bettina Kemme — *Jun 2025 – Dec 2025*

- Instrumented Switch Transformer and Qwen **Mixture-of-Experts** models with CodeCarbon on DeepSpeed to profile GPU energy use; identified optimisations cutting energy by ~**66%** with no loss in accuracy.

McGill University – Prometheus Lab — Montreal, Canada
Undergraduate Researcher · Prof. Joseph Vybihal — *Sep 2024 – Dec 2024*

- Built Python bots to collect 400+ TikTok videos and quantified algorithmic bias in engagement by creator gender and content sentiment (Pandas, scikit-learn).

PROJECTS

Mini-C Compiler | Java, Bash — Jan 2025 – May 2025

- Built a compiler for a subset of C from parsing through code generation, with Bash scripts automating builds and tests.

Paxos Total-Order Multiplayer Game | Java — Nov 2024 – Dec 2024

- Implemented the Paxos consensus algorithm to ensure a consistent game order for all players in a distributed system.

SOC-cessful Schedule | PHP, SQLite, XAMPP — Nov 2024 – Jan 2025

- Designed and built a full-stack booking website used by school staff and students to schedule appointments.

BCV – McWiCS Hackathon | React, Drizzle, PostgreSQL, Gemini API — 2025

- Built a full-stack web app that tailors a CV to a target role by using the Gemini API to order experiences by relevance.

SKILLS

Languages: Python, Java, C/C++, SQL, JavaScript, Bash, OCaml
ML, Data & Frameworks: PyTorch, TensorFlow, scikit-learn, LangChain, Pandas, NumPy, Django, Flask, SQLite
Systems & Tools: AWS (Lambda, Bedrock, S3), Docker, Git, Linux, REST APIs, ZooKeeper, TCP/IP, React
Languages & Interests: English (native), French (native) · Dragon boat, figure skating, dance, piano